

- Nombres Complexes -

① Définition :

- l'ensemble des nombres complexes est noté $\mathbb{C}$ il contient $\mathbb{R}$ ($\mathbb{R} \subset \mathbb{C}$)
- un nombre complexe s'écrit sous sa forme algébrique / cartésienne :

$$z = x + iy \text{ avec } \begin{cases} x = \operatorname{Re}(z) \text{ partie réelle de } z \\ y = \operatorname{Im}(z) \text{ partie imaginaire de } z \end{cases}$$

$$z = 0 \text{ si } x = 0 \text{ et } y = 0$$

Le conjugué d'un nombre complexe s'écrit : $\bar{z} = x - iy$ et ses propriétés sont :

$$\begin{aligned} \overline{z + z'} &= \bar{z} + \bar{z}' \\ \overline{z \cdot z'} &= \bar{z} \cdot \bar{z}' \\ \overline{\frac{z}{z'}} &= \frac{\bar{z}}{\bar{z}'} \end{aligned}$$

$$z \cdot \bar{z} = 2a + 2ib \text{ est imaginaire } \text{eq } \bar{z} = -\bar{z}$$

$$z - \bar{z} = 2ib \text{ est réel } \text{eq } \bar{z} = \bar{z}$$

$$z \cdot \bar{z} = (\operatorname{Re}(z))^2 + (\operatorname{Im}(z))^2$$

$$z = a + ib \quad \bar{z} = a - ib$$

$$\bar{z} = a - ib$$

$$\bar{\bar{z}} = z$$

$$z = -a - ib$$

$$z \cdot z' = (a + ib)(a' + ib')$$

$$z \cdot \bar{z} = (a - ib)(a + ib)$$

$$z \cdot z' = (aa' - bb') + i(ab' + a'b)$$

$$z \cdot \bar{z} = (a - ib)(a + ib)$$

$$z^2 = (z \cdot z) = (a + ib)^2$$

$$z^2 = a^2 - b^2 + 2iab$$

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③ Forme trigonométrique :

$$z = |z|(\cos \theta + i \sin \theta)$$

$$|z| = \sqrt{x^2 + y^2} = \sqrt{z \bar{z}}$$

$$\theta = \arg(z) \text{ est un arg de } z \text{ (définie à } 2k\pi \text{ près)}$$

$$\text{Si } z \neq 0 : \cos \theta = \frac{x}{|z|} = \frac{x}{\sqrt{x^2 + y^2}}$$

$$\sin \theta = \frac{y}{|z|} = \frac{y}{\sqrt{x^2 + y^2}}$$

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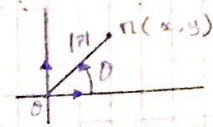
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Propriétés :

$$|z^n| = |z|^n \text{ avec } n \in \mathbb{N}$$

$$|z \cdot z'| = |z| \cdot |z'|$$

$$|\frac{z}{z'}| = \frac{|z|}{|z'|}$$

$$|\frac{1}{z}| = \frac{1}{|z|}$$

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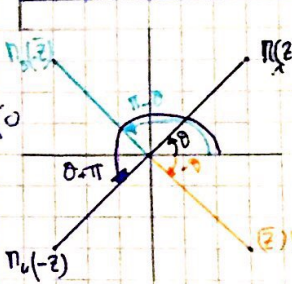
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Forme principale de $(z, \overline{z})$: $z = |z| e^{i\theta}$

z	$1+i$	$1+i\sqrt{3}$	$\sqrt{3}+i$
$\arg(z)$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{6}$



$$\arg(z \cdot z') = \arg(z) + \arg(z')$$

$$\arg(\frac{z}{z'}) = \arg(z) - \arg(z')$$

$$\arg(\frac{1}{z}) = -\arg$$

① Ecriture exponentielle :

Tout nombre complexe $z = re^{i\theta}$ avec $r = |z|$ et $\arg(z) = \theta [2\pi]$ avec $r > 0$
 $|e^{i\theta}| = 1, e^{i\theta} = e^{i(\theta+2\pi k)}$ $\forall k \in \mathbb{Z}$

Equation de second degré dans $\mathbb{C}$
 $az^2 + bz + c = 0$
 $\Delta = b^2 - 4ac$
 $z' = \frac{-b \pm \sqrt{\Delta}}{2a}$ et $z'' = \frac{-b \pm i\sqrt{\Delta}}{2a}$
 $az^2 + bz + c = a(z - z')(z - z'')$
 $z' + z'' = -\frac{b}{a}$ et $z' \cdot z'' = \frac{c}{a}$

$$\begin{aligned} e^{i0} &= e^{-i0} \\ e^{i0} &= e^{i(\theta+\pi)} \\ e^{i\theta} &= e^{i(\theta+\pi)} \\ \frac{e^{i\theta}}{e^{i\theta}} &= e^{i0} \\ \frac{e^{i\theta}}{e^{i0}} &= e^{i(\theta-0)} \end{aligned}$$

$$\begin{aligned} e^{i0} &= 1 \\ e^{i\pi} &= -1 \\ e^{i\pi/2} &= i \\ e^{-i\pi/2} &= -i \end{aligned}$$

Formule d'Euler

$$\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2} \Rightarrow e^{i\theta} + e^{-i\theta} = 2 \cos(\theta)$$

$$\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i} \Rightarrow e^{i\theta} - e^{-i\theta} = 2i \sin(\theta)$$

$$\begin{aligned} 1 + e^{i\theta} &= e^{i\theta/2} + e^{i\theta/2} \\ &= e^{i\theta/2} (e^{-i\theta/2} + e^{i\theta/2}) \\ &= 2 \cos \frac{\theta}{2} e^{i\theta/2} \end{aligned}$$

$$\begin{aligned} 1 - e^{i\theta} &= e^{i\theta/2} - e^{i\theta/2} \\ &= e^{i\theta/2} (e^{-i\theta/2} - e^{i\theta/2}) \\ &= 2i \sin \frac{\theta}{2} e^{i\theta/2} \end{aligned}$$

$$\begin{aligned} e^{i\theta} + e^{i\phi} &= e^{i\frac{\theta+\phi}{2}} \left[\frac{e^{i\frac{\theta-\phi}{2}} + e^{-i\frac{\theta-\phi}{2}}}{2} \right] \\ &= e^{i\frac{\theta+\phi}{2}} \cos \frac{\theta-\phi}{2} \\ e^{i\theta} - e^{i\phi} &= 2i \cos \frac{\theta+\phi}{2} e^{i\frac{\theta-\phi}{2}} \end{aligned}$$

L'ensemble des points Π :

- $AN = l \Rightarrow$ Cercle de centre A et de rayon l
- $AN = AB \Rightarrow$ Cercle de centre A et de rayon AB ou CD
- $AN \perp NB \Rightarrow$ Cercle de diamètre [AB]
- $AN = BN \Rightarrow$ médiatrice de [AB]
- AN et BN col $\Rightarrow$ droite (AB)
- $AN \perp AB \Rightarrow$ droite $\perp$ (AB) passant par A
- $D: ax + by + c = 0 \Rightarrow$ équation d'une droite (D)
- $C: (x-a)^2 + (y-b)^2 = R^2 \Rightarrow$ cercle de centre $M(-a, -b)$ et de rayon R

